

# Network Device Compliance Report

Generated 2026-09-27 13:24

## Device Identification

|                              |                                                                          |
|------------------------------|--------------------------------------------------------------------------|
| Device (from filename)       | cisco_sample.cfg                                                         |
| Hostname                     | EDGE-RTR-01                                                              |
| Detected Vendor              | Cisco                                                                    |
| Privileged Secret Configured | Yes                                                                      |
| Model / Serial Number        | Collected via 'show version' on live device (out of scope for prototype) |

## Compliance Summary

|                      |                              |
|----------------------|------------------------------|
| Framework            | NIST                         |
| Overall Compliance   | 35.7% (5/14 controls passed) |
| Failures by Severity | High: 5   Medium: 4   Low: 0 |

## Compliance Findings

NIST-AC-1Remote Access Encryption (SSHv2)HighFAIL

**Description:** NIST SP 800-53 AC-17: remote access must use encrypted protocols.  
**Evidence:** ip ssh version 1  
**Source:** config line  
**Detected:** 1  
**Expected:** Remote access must use encryption (SSHv2).  
**Why it matters:** NIST SP 800-53 AC-17 requires encryption for all remote access.  
**Potential attack path:** Remote access unencrypted -> Session intercepted -> Admin compromise  
**Remediation (vendor-specific):**  
ip ssh version 2  
*Meaning: Forces SSH v2 to comply with AC-17's encryption requirement.*

NIST-CM-7Insecure Remote Access DisabledHighFAIL

**Description:** NIST CM-7: restrict functionality to least required. Telnet must be disabled.  
**Evidence:** transport input telnet ssh  
**Source:** config line  
**Detected:** Enabled  
**Expected:** Telnet (non-essential insecure service) must be disabled.  
**Why it matters:** NIST CM-7: least functionality - Telnet must be disabled.  
**Potential attack path:** Telnet enabled -> Sniffing yields credentials -> Unauthorized access  
**Remediation (vendor-specific):**  
line vty 0 4  
transport input ssh  
*Meaning: Removes Telnet from the vty lines, complying with least-functionality.*

NIST-SC-8Encrypted Management ChannelMediumPASS

**Description:** NIST SC-8: transmission confidentiality. HTTP management is unencrypted.  
**Evidence:** no ip http server  
**Source:** config line  
**Detected:** Disabled  
**Expected:** Management traffic must be encrypted.

NIST-SC-8Secure Management Channel AvailableMediumFAIL

**Description:** NIST SC-8: encrypted management must be available.  
**Evidence:** No https mgmt enabled configuration detected.  
**Source:** not present in config

**Detected:** Not configured / not detected in this configuration.  
**Expected:** Encrypted HTTPS management must be available.  
**Why it matters:** NIST SC-8: encrypted management must be available.  
**Potential attack path:** No HTTPS available -> Admins use insecure alternatives -> Credential exposure  
**Remediation (vendor-specific):**  
ip http secure-server  
*Meaning: Enables HTTPS management as the encrypted alternative.*

|                                                      |        |      |
|------------------------------------------------------|--------|------|
| NIST-IA-5 Authenticator Management - Password Length | Medium | FAIL |
|------------------------------------------------------|--------|------|

**Description:** NIST IA-5: enforce minimum password complexity.  
**Evidence:** No min password length configuration detected.  
**Source:** not present in config  
**Detected:** Not configured / not detected in this configuration.  
**Expected:** Minimum password length must be at least 8.  
**Why it matters:** NIST IA-5: minimum password complexity required.  
**Potential attack path:** Weak password policy -> Brute-force succeeds -> Account compromise  
**Remediation (vendor-specific):**  
security passwords min-length 8  
*Meaning: Enforces 8-character minimum passwords, complying with IA-5.*

|                                        |      |      |
|----------------------------------------|------|------|
| NIST-IA-5b Password Encryption at Rest | High | PASS |
|----------------------------------------|------|------|

**Description:** NIST IA-5: protect stored authenticators cryptographically.  
**Evidence:** service password-encryption  
**Source:** config line  
**Detected:** Enabled  
**Expected:** Stored authenticators must be cryptographically protected.

|                                        |        |      |
|----------------------------------------|--------|------|
| NIST-AC-11 Session Lock / Idle Timeout | Medium | FAIL |
|----------------------------------------|--------|------|

**Description:** NIST AC-11: sessions must lock after defined inactivity.  
**Evidence:** exec-timeout 30 0  
**Source:** config line  
**Detected:** 30  
**Expected:** Sessions must lock after 10 minutes of inactivity.  
**Why it matters:** NIST AC-11: sessions must lock after inactivity.  
**Potential attack path:** Idle session persists -> Session hijack -> Unauthorized changes  
**Remediation (vendor-specific):**  
line vty 0 4  
exec-timeout 10 0  
*Meaning: Auto-locks idle sessions after 10 minutes, complying with AC-11.*

|                                 |      |      |
|---------------------------------|------|------|
| NIST-AU-2 Audit Logging Enabled | High | FAIL |
|---------------------------------|------|------|

**Description:** NIST AU-2: audit events must be generated and retained.  
**Evidence:** No logging enabled configuration detected.  
**Source:** not present in config  
**Detected:** Not configured / not detected in this configuration.  
**Expected:** Audit events must be generated and retained.  
**Why it matters:** NIST AU-2: audit events must be generated and retained.  
**Potential attack path:** No audit trail -> Attack goes undetected -> No forensic evidence  
**Remediation (vendor-specific):**  
logging host  
logging trap informational  
*Meaning: Sends audit logs to a central server, complying with AU-2.*

|                                |        |      |
|--------------------------------|--------|------|
| NIST-AU-8 Time Synchronization | Medium | PASS |
|--------------------------------|--------|------|

**Description:** NIST AU-8: reliable timestamps require accurate time source.  
**Evidence:** ntp server 10.0.0.1  
**Source:** config line  
**Detected:** Enabled  
**Expected:** Time synchronization must be configured.

|                                               |        |      |
|-----------------------------------------------|--------|------|
| NIST-AC-7 Unsuccessful Login Attempts Limited | Medium | FAIL |
|-----------------------------------------------|--------|------|

**Description:** NIST AC-7: limit consecutive invalid login attempts.  
**Evidence:** No login lockout configured configuration detected.  
**Source:** not present in config  
**Detected:** Not configured / not detected in this configuration.  
**Expected:** Consecutive invalid login attempts must be limited.

**Why it matters:** NIST AC-7: limit unsuccessful login attempts.

**Potential attack path:** Unlimited attempts -> Brute-force succeeds -> Account compromised

**Remediation (vendor-specific):**

login block-for 120 attempts 3 within 60

*Meaning: Locks the account after 3 failed login attempts (AC-7).*

#### NIST-SI-4 No Weak Default SNMP Communities

High

FAIL

**Description:** NIST SI-4: monitoring channels must not use default credentials.

**Evidence:** snmp-server community public RO

**Source:** config line

**Detected:** Enabled

**Expected:** Default SNMP community strings must not be used.

**Why it matters:** NIST SI-4: monitoring must not use default credentials.

**Potential attack path:** Default SNMP string -> Reconnaissance succeeds -> Targeted exploitation

**Remediation (vendor-specific):**

no snmp-server community public

snmp-server community RO

*Meaning: Replaces default SNMP community strings with strong ones (SI-4).*

#### NIST-AC-4 Management Access Restricted

High

FAIL

**Description:** NIST AC-4: information flow control - management interfaces must be restricted.

**Evidence:** No vty acl applied configuration detected.

**Source:** not present in config

**Detected:** Not configured / not detected in this configuration.

**Expected:** Management access must be restricted to trusted sources.

**Why it matters:** NIST AC-4: management access must be restricted.

**Potential attack path:** Management port exposed -> Reachable from attacker segment -> Compromise

**Remediation (vendor-specific):**

access-list 10 permit

line vty 0 4

access-class 10 in

*Meaning: Applies an ACL restricting management to trusted source IPs (AC-4).*

#### NIST-IA-2 Centralized Authentication

Medium

PASS

**Description:** NIST IA-2: identification and authentication - centralized AAA preferred.

**Evidence:** aaa new-model

**Source:** config line

**Detected:** Enabled

**Expected:** Centralized authentication must be configured.

#### NIST-AC-8 System Use Notification (Banner)

Low

PASS

**Description:** NIST AC-8: system use notification before login.

**Evidence:** banner motd ^C Authorized access only. Unauthorized use is prohibited. ^C

**Source:** config line

**Detected:** Enabled

**Expected:** A system use notification must be displayed before login.